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Algo Assignment #1

Q2: $n = \frac{n}{2} = \frac{30}{2} = 15$

$n = \frac{15}{2} = 7$

n is decreasing $n, \frac{n}{2}, \frac{n}{4}, \frac{n}{8}$

$\log_2 n \rightarrow O(\log n)$

Q4: $O(1)$, one statement

Q6: $O(n)$ → j always 0 in inner loop

Q8: $O(n)$, $O(1) \rightarrow O(n)$

Q10: $O(n)$

$O(n * x)$

$O(n) \rightarrow$ ~~$O(2n) \rightarrow O(n)$~~ ~~$O(n^2)$~~ → Function Loop called for ~~every~~ x (undefined) in every n

Q12: l1: 1-n

l2: $1-n^2$

l3: 0-j

Total: $O(n^5)$

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Q14: T : $O(n)$ L1 : n

↓ L2 : n

T : $O(n)$ → $O(n^2)$

Q16: T : $O(\log_2 n)$ i + 2

↓

T : $O(n)$ j++ → $O(n \log n)$

Q18: T : $O(n)$ i to n

T : $O(n)$ → $O(n^2)$

Q20: L1 → T : $O(n)$

L2 → $O(n)$

→ $O(n^2)$

Q22: L1 → $O(n^2)$

Total : $O(n^2)$

Q 24: T : $O(n^2)$

Q26: Insertion Sort

First loop runs $O(n)$

Second loop runs $O(n)$ also in worst case

Total : $O(n^2)$

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Q28: Outer : $O(n)$
Inner : $O(n)$
Total : $O(n^2)$

Q30: L1 : $O(n)$
L2 : $O(n^2)$
L3 : Conditional $\rightarrow O(n^2)$
Total : $O(n^5)$

Q32: L1 : $O(n)$
L2 : $O(31) \rightarrow O(1)$
Total : $O(n)$

Q34: L1 : $O(n)$
L2 : $O(\sqrt{995}) \rightarrow O(1)$
Total : $O(n)$

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Q2.

1. $T(n) = 3T(\frac{n}{4}) + n \quad , T(1) = 1$

$$T(\frac{n}{4}) = 3T(\frac{n}{16}) + \frac{n}{4}$$

$$T(n) = 3\left(3T(\frac{n}{4^2}) + \frac{n}{4}\right) + n$$

$$= 3^2 T\left(\frac{n}{4^2}\right) + \frac{n}{4}) + n$$

$$= 3^2 T\ \frac{n}{4^2} + n\left(1 + \frac{3}{4}^2\right)$$

this goes on till

$$= 3^k T\left(\frac{n}{4^k}\right) + n \sum_{i=0}^{k-1} \left(\frac{3}{4}\right) i \quad \text{— ①}$$

when bare case

$$\frac{n}{4^k} = 1$$

$$\log_4 n = k$$

put in ①

$$3^{\log_4 n}\left(\frac{n}{4^{\log_4 n}}\right) + n \sum_{i=0}^{\log_4 n - 1} \left(\frac{3}{4}\right) i$$

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let $n = 16$

$\log_4 16 = 2$

$4^{\log_4 16} = 4^2 = 16 = n$

so,

$3^{\log_4 n}\left(\frac{n}{n}\right) = 3^{\log_4 n}$

or $= n^{\log_4 3}$

$O(n^{0.79})$

so $TC = O(n)$

Q2. 2.

$T(n) = 2 + T(n-1) \quad T(0) = 2$

$T(n-1) = 2 + T((n-1)-1)$

$= 2 + T(n-2)$

$T(n) = 2 \times 2 + T(n-2)$

this goes on until

$T(n) = 2k + T(n-k)$

so

$T(n) = 2n + T(0)$

$T(n) = 2n + 2$

$T(n) = O(n)$

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Q2 3.

$T(n) = T(n-1) + 1 \qquad T(1) = O(1)$

$T(n-1) = T(n-1-1) + 1$

$= T(n-2) + 1$

$T(n) = d + T(n-2)$

goes on till

$T(n) = k + T(n-k)$

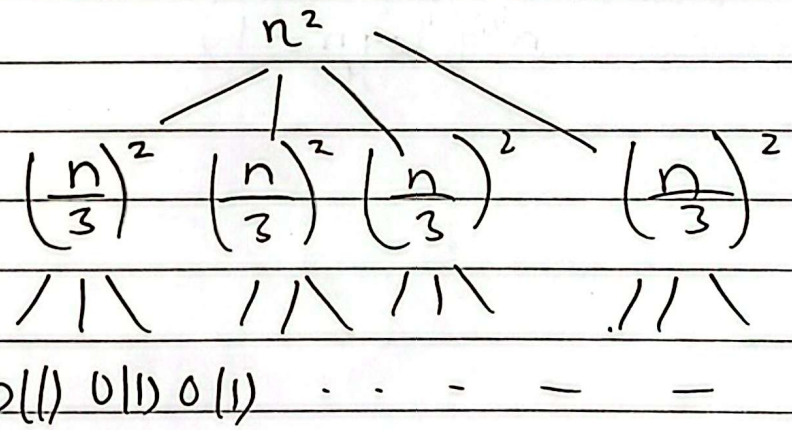
$k = n-1$

$= (n-1) + T(1)$

$T(n) = (n-1) + k$

$T(n) = O(n)$

Q(3)(i) $T(n) = 4T(n/3) + \Theta(n^2)$



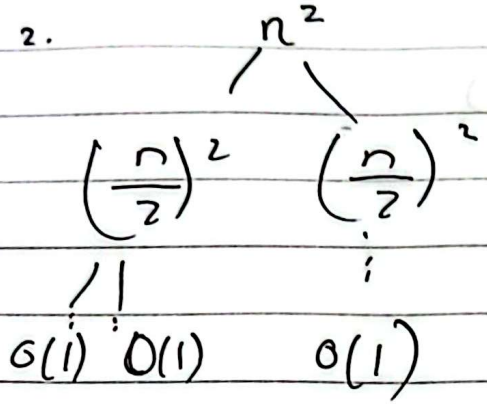
$\frac{n}{3^k} = 1 \qquad n^2\left(1 + \frac{4}{9} + \frac{16}{81} \ldots\right)$

$k = \log_3 n \qquad O(n^2)$

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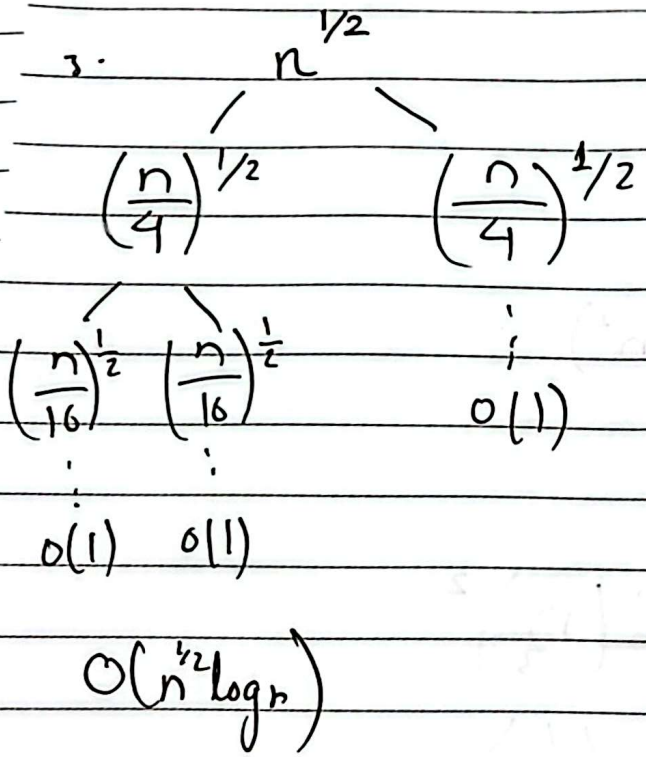
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2.



$O(n^2)$

3.



$O(n^{1/2}\log n)$

$\frac{n}{4^k} = 1$

$k = \log_4 n$

$n^{1/2} + n^{1/2} + \ldots + n^{1/2}$

geometric

$n^{1/2}(\log_4 n + 1)$